MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics **WEEKLY CHAPTER TEST – 02 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 6: Triangles** **MM:** 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. If in two triangles ABC and DEF, $\frac{AB}{DE} = \frac{BC}{FD}$, then they will be similar when: [1]

(a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$

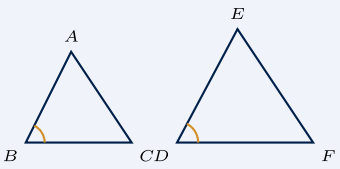
▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) $\angle B = \angle D$

Given: In $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{FD}$.

We know the SAS (Side-Angle-Side) similarity criterion:

Formula / Theorem Used: If one angle of a triangle equals one angle of another triangle and the sides *including* these angles are proportional, the two triangles are similar.



In $\triangle ABC$, the angle included between sides AB and BC is $\angle B$.
In $\triangle DEF$, the angle included between sides DE and FD (i.e., DF) is $\angle D$.
Hence, for SAS similarity ($\triangle ABC \sim \triangle EDF$), we require $\boxed{\angle B = \angle D}$.

Q2. If $\triangle ABC \sim \triangle EDF$ and $\triangle ABC$ is not similar to $\triangle DEF$, then which of the following is **NOT** true? [1]

(a) $BC \cdot EF = AC \cdot FD$ (b) $AB \cdot EF = AC \cdot DE$ (c) $BC \cdot DE = AB \cdot EF$ (d) $BC \cdot DE = AB \cdot FD$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) $BC \cdot DE = AB \cdot EF$

Given: $\triangle ABC \sim \triangle EDF$ (vertex correspondence: $A \leftrightarrow E$, $B \leftrightarrow D$, $C \leftrightarrow F$).

We know that corresponding sides of similar triangles are proportional:

Formula / Theorem Used: $\dfrac{AB}{ED} = \dfrac{BC}{DF} = \dfrac{AC}{EF}$

Checking each pairwise cross-product from the three true proportions:

- From $\dfrac{BC}{DF} = \dfrac{AC}{EF} \implies BC \cdot EF = AC \cdot FD$ $\Rightarrow$ Option (a) is **true**.
- From $\dfrac{AB}{ED} = \dfrac{AC}{EF} \implies AB \cdot EF = AC \cdot DE$ $\Rightarrow$ Option (b) is **true**.
- From $\dfrac{AB}{ED} = \dfrac{BC}{DF} \implies BC \cdot DE = AB \cdot FD$ $\Rightarrow$ Option (d) is **true**.

Option (c) claims $BC \cdot DE = AB \cdot EF \implies \dfrac{BC}{EF} = \dfrac{AB}{DE}$, which together with $\dfrac{AB}{DE} = \dfrac{BC}{DF}$ forces $EF = DF$ (and hence $\triangle ABC \sim \triangle DEF$, contradicting the given condition). Thus, **(c)** is **NOT** true.

Q3. If $\triangle ABC \sim \triangle PQR$ with $\dfrac{BC}{QR} = \dfrac{1}{3}$, then the ratio $\dfrac{\text{Perimeter}(\triangle PQR)}{\text{Perimeter}(\triangle ABC)}$ is: [1]

(a) 9 (b) 3 (c) $\dfrac{1}{3}$ (d) $\dfrac{1}{9}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 3

Given: $\triangle ABC \sim \triangle PQR$ and $\dfrac{BC}{QR} = \dfrac{1}{3}$.

We know the perimeter ratio theorem for similar triangles:

Formula / Theorem Used: If $\triangle ABC \sim \triangle PQR$, then $\dfrac{\text{Perimeter}(\triangle PQR)}{\text{Perimeter}(\triangle ABC)} = \dfrac{PQ}{AB} = \dfrac{QR}{BC} = \dfrac{PR}{AC}$.

Taking the reciprocal of the given ratio $\dfrac{BC}{QR} = \dfrac{1}{3}$:

$$\frac{QR}{BC} = \frac{3}{1} = 3 \implies \frac{\text{Perimeter}(\triangle PQR)}{\text{Perimeter}(\triangle ABC)} = \frac{QR}{BC} = \boxed{3}$$

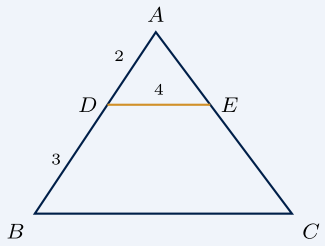
Q4. In $\triangle ABC$, D and E are points on sides AB and AC such that $DE \parallel BC$. If $AD = 2$ cm, $DB = 3$ cm, and $DE = 4$ cm, then the length of BC is: [1]

(a) 6 cm (b) 8 cm (c) 10 cm (d) 12 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) 10 cm

Given: $DE \parallel BC$, $AD = 2$ cm, $DB = 3$ cm, $DE = 4$ cm.
Total side length $AB = AD + DB = 2 + 3 = 5$ cm.
Since $DE \parallel BC$, $\angle ADE = \angle B$ (corresponding angles) and $\angle A$ is common, so $\triangle ADE \sim \triangle ABC$ (AA similarity):



Formula / Theorem Used: $\triangle ADE \sim \triangle ABC \implies \dfrac{AD}{AB} = \dfrac{DE}{BC}$

Substituting $AD = 2$, $AB = 5$, and $DE = 4$:

$$\frac{2}{5} = \frac{4}{BC} \implies 2 \times BC = 4 \times 5 = 20 \implies BC = \frac{20}{2} = \boxed{\mathbf{10}\text{ cm}}$$

Q5. Assertion (A): All equilateral triangles are similar to each other.
Reason (R): All equilateral triangles have each interior angle equal to 60°, satisfying the AAA similarity criterion. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

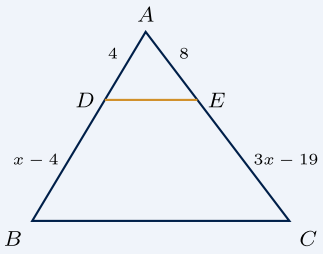
We know the AAA similarity criterion for triangles:

Formula / Theorem Used: If all three corresponding angles of two triangles are equal, then the triangles are similar.

Every equilateral triangle has three equal sides and therefore three equal interior angles of $\frac{180^\circ}{3} = 60^\circ$ each. Thus, any two equilateral triangles are equiangular $(60^\circ, 60^\circ, 60^\circ)$ and hence similar by AAA similarity. Both (A) and (R) are true, and (R) correctly explains (A).

Q6. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 4$ cm, $DB = (x - 4)$ cm, $AE = 8$ cm, and $EC = (3x - 19)$ cm. Find the value of x. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Applying Basic Proportionality Theorem [1 Mark]

Given: In $\triangle ABC$, $DE \parallel BC$, $AD = 4$ cm, $DB = (x - 4)$ cm, $AE = 8$ cm, and $EC = (3x - 19)$ cm.

We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: In $\triangle ABC$, if $DE \parallel BC$, then $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.

Substituting the given values:

$$\frac{4}{x - 4} = \frac{8}{3x - 19}$$

Step 2: Solving the Equation for x [1 Mark]

Cross-multiplying both sides:

$$4(3x - 19) = 8(x - 4)$$

Expanding brackets on both sides:

$$12x - 76 = 8x - 32$$

Transposing $8x$ to the LHS and -76 to the RHS:

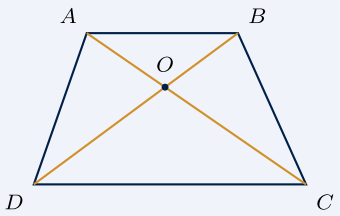
$$12x - 8x = 76 - 32 \implies 4x = 44 \implies x = \frac{44}{4} = 11$$

Verifying positive segment lengths: $DB = 11 - 4 = 7$ cm > 0 and $EC = 3(11) - 19 = 14$ cm > 0.

$$\boxed{x = 11}$$

Q7. The diagonals of a quadrilateral $ABCD$ intersect each other at the point O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show that $ABCD$ is a trapezium. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: A quadrilateral $ABCD$ whose diagonals AC and BD intersect at O such that $\frac{AO}{BO} = \frac{CO}{DO}$.
To Prove: $ABCD$ is a trapezium (i.e., $AB \parallel DC$).
We know the SAS similarity criterion for triangles:

Formula / Theorem Used: If two pairs of sides are proportional and their included angles are equal, the triangles are similar, so all corresponding angles are equal.

Step 1: Rearranging the Proportion & Establishing Similarity [1.5 Marks]

From the given relation $\frac{AO}{BO} = \frac{CO}{DO}$, interchanging BO and CO (alternendo):

$$\frac{AO}{CO} = \frac{BO}{DO}$$

In $\triangle AOB$ and $\triangle COD$:

1. $\frac{AO}{CO} = \frac{BO}{DO}$ (Proved above)
2. $\angle AOB = \angle COD$ (Vertically opposite angles)

By the **SAS similarity criterion**, $\triangle AOB \sim \triangle COD$.

Step 2: Proving Parallelism of Opposite Sides [1.5 Marks]
Since corresponding angles of similar triangles are equal:

$$\angle OAB = \angle OCD \quad (\text{i.e., } \angle CAB = \angle ACD)$$

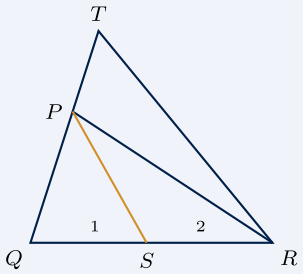
These are alternate interior angles formed by lines AB and DC with transversal AC. Since alternate interior angles are equal, $AB \parallel DC$.
Because one pair of opposite sides of quadrilateral $ABCD$ is parallel:

$\therefore ABCD$ is a trapezium. [Hence Proved]

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Q8. In the given figure, $\frac{QT}{PR} = \frac{QR}{QS}$ and $\angle 1 = \angle 2$. Prove mathematically that $\triangle PQS \sim \triangle TQR$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: $\frac{QT}{PR} = \frac{QR}{QS}$ and $\angle 1 = \angle 2$ (where $\angle 1 = \angle PQR$ and $\angle 2 = \angle PRQ$).
To Prove: $\triangle PQS \sim \triangle TQR$.

We know the isosceles triangle property and the SAS similarity criterion:

Formula / Theorem Used: Sides opposite equal angles of a triangle are equal. If two sides of one triangle are proportional to two sides of another and the included angles are equal, the triangles are similar (SAS).

Step 1: Using $\angle 1 = \angle 2$ in $\triangle PQR$ [1 Mark]
In $\triangle PQR$, $\angle PQR = \angle PRQ$ ($\angle 1 = \angle 2$). Since sides opposite equal angles are equal:

$$PR = PQ \qquad \text{— (i)}$$

Step 2: Substituting $PR = PQ$ into the Given Ratio [1 Mark]
We are given $\frac{QT}{PR} = \frac{QR}{QS}$. Replacing PR with PQ from (i):

$$\frac{QT}{PQ} = \frac{QR}{QS}$$

Taking reciprocals on both sides:

$$\frac{PQ}{QT} = \frac{QS}{QR} \qquad \text{— (ii)}$$

Step 3: Applying SAS Similarity to $\triangle PQS$ and $\triangle TQR$ [1 Mark]
In $\triangle PQS$ and $\triangle TQR$:

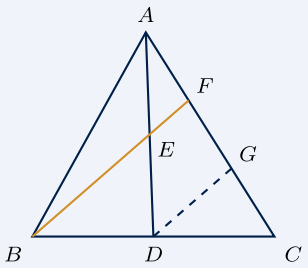
1. $\frac{PQ}{TQ} = \frac{QS}{QR}$ (from (ii))
2. $\angle PQS = \angle TQR$ ($= \angle 1$, common angle included between the proportional sides)

$\therefore \triangle PQS \sim \triangle TQR$ (by SAS similarity) [Hence Proved]

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Q9. In $\triangle ABC$, D is the midpoint of BC and E is the midpoint of AD. If BE produced meets AC at F, prove that $AF = \frac{1}{3}AC$. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: In $\triangle ABC$, D is the midpoint of BC ($BD = DC$) and E is the midpoint of AD ($AE = ED$). Line BE produced meets AC at F.
To Prove: $AF = \frac{1}{3}AC$.
Construction: Draw $DG \parallel BF$ meeting AC at point G.
We know the Basic Proportionality Theorem (Thales' Theorem):

> **Formula / Theorem Used:** If a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides those two sides in the same ratio.

Step 1: Applying BPT in $\triangle ADG$ [1 Mark]
In $\triangle ADG$, $EF \parallel DG$ (since $DG \parallel BF$). By BPT:

$$\frac{AF}{FG} = \frac{AE}{ED}$$

Since E is the midpoint of AD, $AE = ED \implies \frac{AE}{ED} = 1$:

$$\frac{AF}{FG} = 1 \implies AF = FG \qquad \text{— (i)}$$

Step 2: Applying BPT in $\triangle CBF$ [1 Mark]
In $\triangle CBF$, $DG \parallel BF$. By BPT:

$$\frac{FG}{GC} = \frac{BD}{DC}$$

Since D is the midpoint of BC, $BD = DC \implies \frac{BD}{DC} = 1$:

$$\frac{FG}{GC} = 1 \implies FG = GC \qquad \text{— (ii)}$$

Step 3: Combining Relations for AC [1 Mark]
From (i) and (ii), $AF = FG = GC$. Since $AC = AF + FG + GC$:

$$AC = AF + AF + AF = 3AF$$

Dividing both sides by 3:

$$\boxed{\therefore AF = \frac{1}{3}AC \quad \text{[Hence Proved]}}$$

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Q10. Case Study 02: Architectural Scale Model & Farmhouse Roof Pyramid

A student prepares a scale model of a farmhouse roof for an exhibition. The attic floor, $ABCD$, is a square of side 12 m. The roof beams form a pyramid with apex T, where each lateral edge AT, BT, CT, DT measures 12 m. Supporting horizontal beams form a square $EFGH$ where E, F, G, H are the midpoints of AT, BT, CT, DT respectively. [4]

(i) **[1 Mark]** State the similarity criterion between $\triangle TEF$ and $\triangle TAB$.

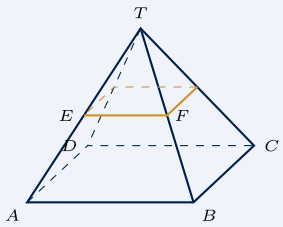
(ii) **[1 Mark]** Find the length of the horizontal supporting beam EF.

(iii) **[2 Marks]** What is the ratio of the perimeter of square $EFGH$ to the perimeter of square $ABCD$? Justify your answer.

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given Data: Square base $ABCD$ has side $AB = BC = CD = DA = 12$ m. Apex is T, lateral edges $AT = BT = CT = DT = 12$ m, and E, F, G, H are the midpoints of AT, BT, CT, DT respectively.

We know the SAS similarity criterion and perimeter scaling property:



Formula / Theorem Used: If two pairs of sides are in the same ratio and their included angles are equal, then the triangles are similar (SAS).

Sub-part (i): Similarity Criterion Between $\triangle TEF$ and $\triangle TAB$ [1 Mark]

Since E and F are the midpoints of AT and BT respectively:

$$TE = \frac{1}{2}TA \implies \frac{TE}{TA} = \frac{1}{2} \quad \text{and} \quad TF = \frac{1}{2}TB \implies \frac{TF}{TB} = \frac{1}{2}$$

Thus, in $\triangle TEF$ and $\triangle TAB$: $\dfrac{TE}{TA} = \dfrac{TF}{TB} = \dfrac{1}{2}$, and $\angle ETF = \angle ATB$ (common included angle).

$$\boxed{\triangle TEF \sim \triangle TAB \quad \text{by the } \textbf{SAS Similarity Criterion.}}$$

Sub-part (ii): Length of Horizontal Supporting Beam EF [1 Mark]

Since $\triangle TEF \sim \triangle TAB$, their corresponding sides are proportional:

$$\frac{EF}{AB} = \frac{TE}{TA} = \frac{1}{2} \implies \frac{EF}{12} = \frac{1}{2} \implies EF = \frac{12}{2} = 6 \text{ m} \implies \boxed{EF = \textbf{6} \text{ m}}$$

Sub-part (iii): Ratio of Perimeters of Square $EFGH$ and Square $ABCD$ [2 Marks]

Applying the same SAS similarity to faces $\triangle TBC$, $\triangle TCD$, and $\triangle TDA$, each side of square $EFGH$ equals half the corresponding side of square $ABCD$, so $EF = FG = GH = HE = \frac{1}{2}(12) = 6$ m.

Perimeter$(EFGH) = 4 \times 6 = 24$ m and Perimeter$(ABCD) = 4 \times 12 = 48$ m.

$$\frac{\text{Perimeter}(EFGH)}{\text{Perimeter}(ABCD)} = \frac{24}{48} = \frac{1}{2} \implies \boxed{\text{Required Ratio } = \textbf{1 : 2}}$$

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